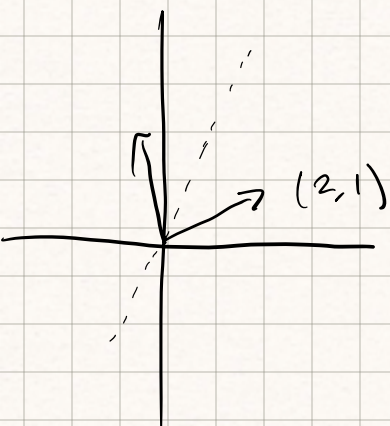


$$\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & -1/5 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}}_{//}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & -1/5 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1/5 & 2/5 \\ -2/5 & 1/5 \end{bmatrix} = \begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix}$$



$$\begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 11/5 \end{bmatrix}$$

$$\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} = \frac{1}{2} \frac{1}{5} \frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ = \frac{1}{20} \begin{bmatrix} -3-4\sqrt{3} & 4-3\sqrt{3} \\ -3\sqrt{3}+4 & 4\sqrt{3}+3 \end{bmatrix}$$

Note!

- 1) order of linear transformations matter
- 2) geometric transformations are linear
- 3) eigen bases are useful (need change of bases)

Exam problem 1:

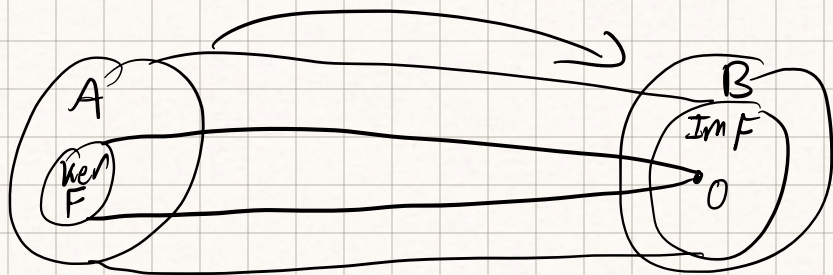
$$\{ v \in \mathbb{R}^3 \mid F(v) = 0 \} \text{ for some linear func } F$$

This is a subspace of $\mathbb{R}^3$

Def: The kernel of $F: A \rightarrow B$ is $\{v \in A \mid F(v) = 0\}$

Lemma: The kernel of a linear transformation is a subspace of the domain

Ex: $\text{Ker}(D) = \{\text{constant functions}\}$



Def: let $f: A \rightarrow B$ the set $\text{Im}(f) = \{b \in B \mid \exists a \in A, f(a) = b\}$ is called the image of f

what can we say about $\dim(\text{Ker } F), \dim(\text{Im } F), \dim A, \dim B$

lemma $\text{Im } F$ is a subspace of B

Ex $F: \mathbb{R}^3 \rightarrow \mathbb{R} \quad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$

$$\dim(A) = 3 \quad \dim(B) = 1 \quad \dim(\text{Im } F) = 0 \quad \dim(\text{Ker } F) = 3$$

Def: The trivial vector space (over any field of scalars) is the vector space w/ 1 element (the 0 vector)

Def: $\dim(\text{trivial}) = 0$

Ex: $F: \mathbb{R}^3 \rightarrow \mathbb{R} \quad \begin{bmatrix} 1 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 10x + 2y + z$

$$\dim(A)=3 \quad \dim(B)=1 \quad \dim(\text{Im} f)=1$$

$$\dim(\ker f) \leq 3$$

if C is a subspace of D then $\dim C \leq \dim D$

$$F: A \rightarrow B \quad \dim(\ker F) \leq \dim A \quad \dim(\text{Im} F) \leq \dim B$$

$$10x + 2y + z = 0 \quad z = -10x - 2y \quad x, y \text{ are free}$$

so $\begin{bmatrix} x \\ y \\ -10x - 2y \end{bmatrix}$ is in $\ker f$ for all x, y can prove this is a basis for $\ker f$

$$= x \begin{bmatrix} 1 \\ 0 \\ -10 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad \ker f = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ -10 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$$

$$\text{so } \dim(\ker f) = 2$$

Thm! Let $F: A \rightarrow B$ be linear and suppose that $\dim A, \dim B$ are finite

$$\dim(\ker F) + \dim(\text{Im} F) = \dim A$$

if $\dim B < \dim A$, F must have a nontrivial kernel

Ex: $F: \mathbb{R}^{2025} \rightarrow \mathbb{R}^{17}$ what is the smallest kernel of F

$$\dim(\ker F) + 17 = 2025 \quad \dim(\ker F) = 2008$$

rank-nullity theorem

$$\underbrace{\dim(\ker F)}_{\text{nullity}} + \underbrace{\dim(\text{Im} F)}_{\text{rank}}$$